

Assessing the Impacts of Infrastructure Development

TRANSPORTATION PLANNING LABORATORY (CE308P)

Group Details

Group Number: 7

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1. Abstract:

This Study evaluates the impact of infrastructure development on transportation networks in 11 selected zones of Chittoor district. The road network was prepared by incorporating centroids, nodes, and centroid connectors. The travel time matrix was estimated using an average speed of 25 km/h. The Trip generation model is calibrated using the ordinary least squared regression method for 66 zones. Trip distribution is estimated using a doubly constrained gravity model.

Traffic assignments were carried out across the network to analyze travel times in multiple scenarios and reduce them on selected links, simulating infrastructure improvements. The performance was evaluated in terms of VHT (vehicle hours travelled) and travel costs. The results indicate that targeted improvements on highly congested links can reduce overall travel time and give economic returns.

2. Introduction and background:

Transportation planning plays an important role in the movement of traffic in urban areas. With an increase in population, there is an increase in travel costs, which often results in congestion and an increase in travel time. Therefore, a systematic data-driven approach in transportation planning is essential for sustainable development.

The transportation planning process includes trip generation, trip distribution, modal split, and traffic assignment. Trip generation estimates the number of trips produced and attracted by zones. Trip distribution estimates the interactions between the zones. Traffic assignments allocate these trips onto the road network to identify congestion.

In this study, the transportation network containing 11 Transportation analysis zones (TAZs) were analyzed. The road network was prepared by incorporating road networks, centroids, nodes, and centroid connectors. The travel time matrix was estimated. Traffic assignment is performed to evaluate network performance and identify congested links. Multiple scenarios are considered by reducing travel times on selected links to simulate network enhancements. The objective of this study is to assess the impact of such developments and to identify the most efficient alternative based on reductions in total vehicle hours travelled and travel costs.

3. Overview of Methodology:

In this study, we have used a systematic approach to analyse travel costs and evaluate the performance of a transportation network. This is followed by the estimation of travel time and traffic assignment. The first step is network preparation with given zones and roads clipped from IND Roads using a built-in QGIS plugin called qaequilibrea. After creating nodes and network, centroid connectors were added and saved as links in the road network. This helps us calculate the cost (C_{ij}) matrix.

As a second step, Trip generation is carried out using an OLS regression model. This method is used to estimate future trip productions and attractions.

In the next stage, Trip distribution is carried out using a doubly constrained gravity model. This estimates the number of trips between zones based on travel time. The resulting origin-destination matrix is then used in traffic assignment to allocate trips on the road network and identify congestion patterns.

To assess the impact of infrastructure improvement, multiple scenarios were considered, reducing travel time on selected links. For each scenario, trip distribution and traffic assignment processes were repeated to account for travel costs. The performance of each scenario was determined based on VHT (vehicle hours travelled) and associated travel costs. This helps to find the most efficient alternative.

4. Study Area description and Network Preparation

This section provides a summary of the study area and preparation of transportation network used for analysis.

4.1 Study Area Description

The study area comprises of 11 Traffic analysis zones located in Chittoor district. These zones include:

1.Chittoor, 2.Palamaner, 3.Baireddipalle, 4.Venkatagirikota, 5. Santhipuram, 6.Gudupalle, 7.Kuppam, 8.Ramakuppam, 9.Bangarupalem, 10.Yadamari and 11.Gudipala. The variation in the population and workforce characteristics plays a very important role in the estimation of trip productions and trip attractions.

4.2 Zoning

Zoning for the study area was predefined as part of the problem statement. The analysis is based on the given 11 Traffic Analysis Zones (TAZs), which are considered as the

fundamental units for estimating trip productions, attractions, and inter-zonal travel patterns.

4.3 Network Preparation

The road network for the study area was prepared using qgis using nodes and links. The existing road network was digitized, and necessary attributes like distance, link type, and direction were added. Centroids were added to each TAZ to represent the origin and destination of trips. These centroids were connected to the nearest nodes using centroid connectors for movement between zones in the network.

The prepared network was then used to compute the shortest path between centroids. A speed of 25 km/h was used to convert distance into travel time. The resulting cost matrix (C_{ij}) serves as a key input for subsequent trip distribution and traffic assignment.

4.4 Cost Matrix (C_{ij})

The cost matrix (C_{ij}) represents the travel time between Traffic analysis zones (TAZs).

From the matrix, it is observed that the travel times vary significantly across zones; this indicates differences in distance and connectivity. Zones with lower travel costs tend to have higher interaction, and those with higher travel costs have fewer interactions. The diagonal elements of the matrix are zero, indicating that there is no intra-zonal travel. The cost matrix is provided in [Appendix A](#).

5. Trip Generation

This section presents the trip generation process, which involves estimating the number of trips produced (O_i) and attracted (D_j) in each Traffic Analysis Zone (TAZ) of the study area. OLS regression is used to establish relationships between socio-economic variables and trip-making behavior.

5.1 Data Used

The trip generation analysis is carried out using the data set given for 66 zones of Chittoor district. The variables in the data set include area, population, and number of workers that influence travel demand.

5.2 Significant Variables

Trip Attraction

From regression analysis, the variable influencing trip attraction is **Total Workers**.

$$A_i = 13.42 + 0.01146(\text{Workers})$$

Table 1: Regression Statistics

Variable	Coefficient	P-Value
Total Workers	0.01	0.00
Constant	13.42	0.06

$R^2 = 0.974$. The model is a strong fit. Total Workers is highly significant, while the constant is not statistically significant.

Trip Production

From regression analysis, the variables influencing trip production are Total Workers and Population.

$$P_i = -23.2468 + 0.0067(\text{Workers}) + 0.0033(\text{Population})$$

Table 2: Regression Statistics

Variable	Coefficient	P-Value
Total Workers	0.0067	0.00
Population	0.0033	0.00
Constant	-23.2468	0.055

$R^2 = 0.980$. The model shows a very strong fit. Both Total Workers and Population are statistically significant ($p < 0.05$), while the constant is slightly insignificant. Multicollinearity may exist due to inclusion of correlated variables such as total workers and its components (male and female workers). The predicted trip production and attractions for 2026 are provided in [Appendix B](#)

5.3 Balancing Productions & Attractions

Since the most accurate model is production, Attractions will be re-estimated using the formula below. $A'_j = A_j \times \frac{\sum P}{\sum A}$

5.4 Final Estimation

Table 3: Corrected attractions.

Zone Id	Zone	Productions	Corrected attractions
1	Chittoor	1194.47	1015.13
2	Palamaner	485.25	478.69
3	Baireddipalle	354.72	384.68
4	Venkatagirikota	560.78	580.22
5	Santhipuram	403.21	444.56
6	Gudupalle	273.01	300.47
7	Kuppam	771.26	781.88

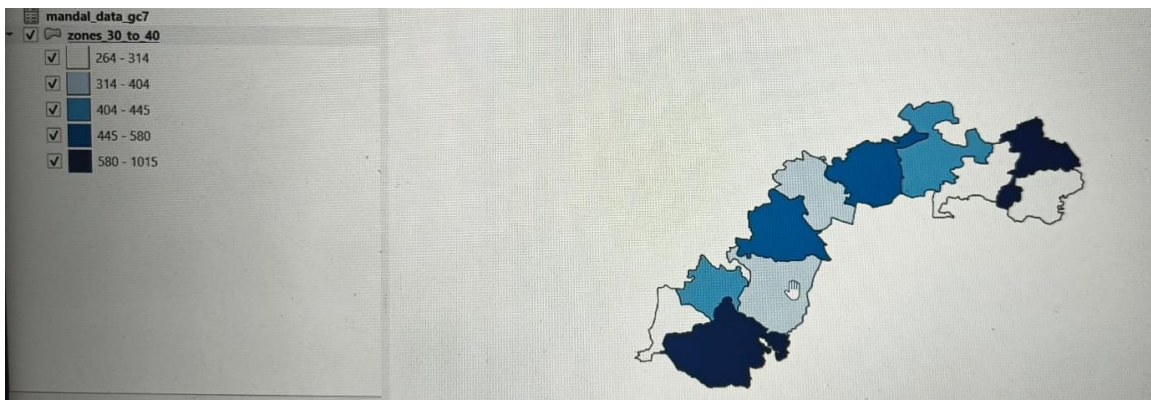
8	Ramakuppam	369.11	403.97
9	Bangarupalem	424.52	414.24
10	Yadamarri	296.9	313.89
11	Gudipala	248.21	263.71
		5381.45	5381.45

5.5 Visualization

Fig 1: Trip Production



Fig 2: Trip Attraction



6. Trip Distribution

This section provides a summary of the trip distribution process, that is used to estimate the number of trips between different Traffic Analysis Zones (TAZs).

6.1 Gravity Model

The Gravity Model is used to estimate the number of trips between different Traffic Analysis Zones (TAZs). The assumption of the gravity model is that the interaction between two zones is directly proportional to the trip productions and attractions, and inversely proportional to the travel cost between them.

In this doubly constrained gravity model, both trip productions (O_i) and trip attractions (D_j) are satisfied. We use an impedance function F_{cij} derived from travel cost matrix.

$$T_{ij} = A_i B_j O_i D_j F_{cij}$$

$$F_{cij} = e^{-0.01 c_{ij}}$$

6.2 Calibration

The impedance function F_{cij} is computed using the given value of β , i.e., 0.01. The balancing factors A_i and B_j are then determined iteratively to ensure that the row totals match trip productions (O_i) and the column totals match trip attractions (D_j).

6.3 Results

The matrix shows the number of trips taken between each pair of zones, satisfying both production and attraction constraints. The results indicate that zones with lower travel costs have higher trip interactions. The total number of trips distributed in the system is 5381.45, which matches the sum of total trip productions and total trip attractions. The complete trip distribution matrix is provided in [Appendix C](#)

6.4 Convergence

The iterations were carried out until convergence was achieved. In this analysis, convergence was achieved after **4 iterations**, with row and column errors within acceptable limits ($\pm 2\%$). This indicates that the model results are reliable.

7. Traffic Assignment

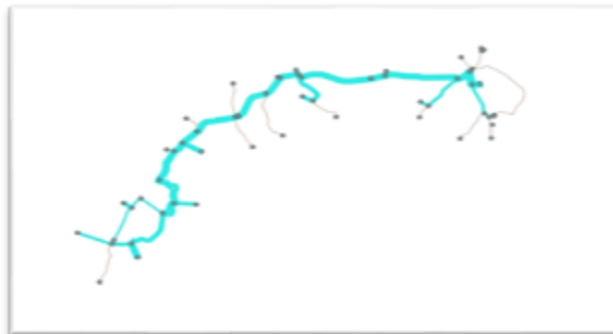
Traffic assignment is carried out to allocate the trips between zones onto the road network and evaluate the network performance. The origin–destination matrix (T_{ij}) is assigned using Aequilibræ based on travel time, and the overall performance is assessed using Vehicle Hours Travelled (VHT).

7.1 Base Case

For the base case, the network is assigned using the initial cost matrix obtained from travel time. A uniform capacity of 2000 is assigned to all links in the network. Traffic assignment is then performed to obtain the flow on each link and evaluate the existing network performance in terms of vehicle hours traveled (VHT)

$$VHT = \sum (T_{ij} \times C_{ij})$$

Fig 3 : Base case scenario traffic analysis



The total base VHT was found to be **7,715.68 veh-hr/day**, representing the existing network condition.

7.2 Selection of Links

Links were selected for improvement based on the following criteria: High traffic flow (PCE flow), High delay factor (ratio of congested time to free-flow time), Significant contribution to total VHT. Based on these criteria, the most critical links were identified and shortlisted for scenario analysis. These links represent the most congested and influential parts of the network. The summary of the attributes of all candidate links and the rationale for each scenario is attached in APPENDIX D.

7.3 Scenario Development

Four improvement scenarios were developed by increasing the capacity (doubling) of selected links:

Scenario 1: Links 7, 21, 29

Scenario 2: Links 7, 21, 40

Scenario 3: Links 29, 40, 25

Scenario 4: Links 33, 35, 40

Scenario datasets were generated, converted to CSV, and used to compute Vehicle Hours Travelled (VHT) for the base and alternative scenarios.

8 Economic analysis

8.1 Cost of Travel

The travel cost calculated using a unit cost of **₹10 per vehicle-hour**: $\text{Cost} = \text{VHT} \times 10$ For the base case, Travel cost = **₹77,156.8 per day**

8.2 Cost of Road Widening

The cost of widening was estimated based on link length and a standard cost per km. Total widening cost for each scenario was calculated as:

$$\text{Cost} = \sum(\text{Length of links} \times \text{Cost per km})$$

8.3 Comparison of Scenarios

1. Scenarios 1 and 2 show very small improvements despite high cost
2. Scenario 3 results in negative impact (increase in VHT)
3. Scenario 4 provides maximum reduction in VHT (~16%)

8.4 Best Alternative

Among all scenarios, **Scenario 4 (Links 33, 35, 40)** is the best alternative because: Highest reduction in VHT, lowest widening cost, Best ROI compared to others, Improves a connected corridor rather than isolated links

Results and discussion

Scenarios 1 and 2: High cost, negligible savings

Although links 7 and 21 have the highest delay factors and VHT contributions, they are 16.30–16.63 km long, making widening costs ₹407–416 Cr each. The dynamic analysis yields only 57–62 veh·hr/day savings ($\leq 0.80\%$ reduction), as the gravity model induces additional trips in response to faster travel times on these long corridors, partially offsetting gains.

Scenario 3: Negative outcome (Braess Paradox)

Widening links 29, 40, and 25 — which lie on three spatially disconnected corridors — increased system VHT by 7.84 veh·hr/day. The induced demand loaded onto unchanged network links increased congestion elsewhere beyond the savings on widened links. This is a manifestation of the Braess Paradox, demonstrating that capacity additions on disconnected links can worsen overall system performance.

Scenario 4: Optimal result

Scenario 4 achieves a VHT reduction of 1,237.98 veh·hr/day (16.04%) at the lowest widening cost (₹549.5 Cr), yielding the highest ROI ratio (0.000822) and shortest payback period (~1,216 years). The strong outcome reflects the spatial coherence of the three links: widening the node-36/node-34 hub creates systemic congestion relief in that corridor, and the re-calibrated gravity model redistributes trips beneficially through the improved sub-network rather than inducing adverse loading elsewhere.

Conclusion

Scenario 4 (links 33, 35, 40) is the recommended alternative. It delivers the largest VHT reduction (16.04%), the lowest investment (₹549.5 Cr), and the best ROI. The analysis demonstrates that effective capacity investment must account for network topology and corridor connectivity. Targeting the most congested individual links (Scenarios 1–2) yields negligible system benefits at enormous cost; widening unconnected links (Scenario 3) worsens the system via induced demand. Only corridor-consistent widening (Scenario 4) achieves meaningful, cost-effective system improvement.

Appendix

Appendix A. Network preparation

TABLE 4 : Cost matrix

From/Tozones	1	2	3	4	5	6	7	8	9	10	11
1	0.00	136.75	158.91	208.09	291.81	325.89	305.21	260.87	69.40	51.48	33.14
2	136.75	0.00	79.00	128.19	211.91	245.99	225.31	180.97	67.73	145.22	161.23
3	158.91	79.00	0.00	49.94	133.66	167.74	147.06	102.72	89.89	167.38	183.39
4	208.09	128.19	49.94	0.00	107.57	141.64	120.96	76.63	139.07	216.56	232.58
5	291.81	211.91	133.66	107.57	0.00	53.45	54.21	54.85	222.80	300.29	316.30
6	325.89	245.99	167.74	141.64	53.45	0.00	40.80	88.93	256.87	334.36	350.38
7	305.21	225.31	147.06	120.96	54.21	40.80	0.00	68.25	236.19	313.69	329.70
8	260.87	180.97	102.72	76.63	54.85	88.93	68.25	0.00	191.86	269.35	285.36
9	69.40	67.73	89.89	139.07	222.80	256.87	236.19	191.86	0.00	77.87	93.89
10	51.48	145.22	167.38	216.56	300.29	334.36	313.69	269.35	77.87	0.00	75.97
11	33.14	161.23	183.39	232.58	316.30	350.38	329.70	285.36	93.89	75.97	0.00

Appendix B. Trip Generation

Table 5: Predicted Trip productions and Trip Attraction and corrected trip attractions

Zone Id	Zone	Productions	Attractions	Corrected attractions
1	Chittoor	1194.47	906.92	1015.13
2	Palamaner	485.25	427.67	478.69
3	Baireddipalle	354.72	343.68	384.68
4	Venkatagirikota	560.78	518.37	580.22
5	Santhipuram	403.21	397.17	444.56
6	Gudupalle	273.01	268.44	300.47
7	Kuppam	771.26	698.53	781.88
8	Ramakuppam	369.11	360.91	403.97
9	Bangarupalem	424.52	370.08	414.24
10	Yadamarri	296.9	280.43	313.89
11	Gudipala	248.21	235.6	263.71
Sum		5381.45	4807.79	5381.45

Appendix C. Trip Distribution

Table 6 : Doubly Constrained gravity model - result (iteration 4)

Zones	1	2	3	4	5	6	7	8	9	10	11	Total Oi
1	549.89	84.97	49.53	47.89	17.12	9.20	25.18	20.60	121.54	128.27	126.94	1181.14
2	63.56	151.34	49.96	48.32	17.27	9.28	25.40	20.78	56.08	22.79	16.00	480.79
3	31.60	42.62	68.32	65.57	23.44	12.60	34.47	28.20	27.88	11.33	7.95	353.98
4	30.88	41.65	66.26	172.64	48.63	26.13	71.51	58.50	27.25	11.08	7.77	562.30
5	9.82	13.24	21.07	43.24	104.72	46.36	102.38	53.41	8.66	3.52	2.47	408.90
6	5.27	7.10	11.30	23.20	46.29	59.69	88.32	28.66	4.65	1.89	1.33	277.70
7	15.84	21.36	33.98	69.75	112.30	97.03	324.68	86.16	13.97	5.68	3.99	784.73
8	12.07	16.28	25.90	53.16	54.59	29.33	80.27	83.40	10.65	4.33	3.04	373.01
9	96.76	59.68	34.79	33.64	12.03	6.46	17.68	14.47	85.69	34.70	24.35	420.24
10	99.22	23.57	13.74	13.28	4.75	2.55	6.98	5.71	33.71	64.80	24.97	293.30
11	100.23	16.89	9.84	9.52	3.40	1.83	5.00	4.09	24.16	25.49	44.89	245.35
Total Dj	1015.13	478.69	384.68	580.22	444.56	300.47	781.88	403.97	414.24	313.89	263.71	5381.45

APPENDIX D – Traffic Assignment

Table 7: Attributes of candidate links selected for widening scenarios

Link	Length (km)	Delay factor	VOC	PCE flow	Cong. time (min)	Widen cost (₹ Cr)	Used in scenario
7	16.63	1.278	0.278	983	51.00	415.8	Sc 1, Sc 2
21	16.30	1.277	0.277	976	49.96	407.4	Sc 1, Sc 2
29	10.27	1.273	0.273	989	31.40	256.9	Sc 1, Sc 3
25	9.78	1.270	0.270	945	29.80	244.4	Sc 3
40	9.66	1.267	0.267	980	29.37	241.6	Sc 2, Sc 3, Sc 4
33	3.14	1.267	0.267	980	9.55	78.5	Sc 4
35	9.18	1.267	0.267	980	27.90	229.4	Sc 4

Table 8: Scenario comparison: VHT savings, travel cost savings, widening cost, and ROI (base VHT = 7,715.68 veh·hr/day; ₹10/veh·hr)

Sc	Strategy	Links	Base VHT	New VHT	Saving (veh·hr)	% Red.	Widen cost (₹ Cr)	ROI ratio
Sc 1	Top 3 delay factor	7,21,29	7,715.68	7,654.12	61.56	0.80%	1,080.1	0.0000208
Sc 2	Top 3 VHT contrib.	7,21,40	7,715.68	7,658.16	57.52	0.75%	1,064.9	0.0000197
Sc 3	Skip long corridors	29,40,25	7,715.68	7,723.52	-7.84	-0.10%	742.8	-0.0000004
Sc 4	Corridor cluster	33,35,40	7,715.68	6,477.70	1,237.98	16.04%	549.5	0.000822

Code For Scenario analysis:

```
import openmatrix as omx
import numpy as np
import os
import pandas as pd
FOLDER = r'C:\Users\cheta\OneDrive\Desktop\lab_project_1\new_project\matrices'
TRIPS_FILE = os.path.join(FOLDER, 'tj_1.omx') # base OD trip matrix
BASE_VHT = 7715.68 # your base VHT (veh·hr/day)
VHT_COST_RS = 10 # ₹ per vehicle-hour
WIDEN_COST_CR = 250000000 # ₹ Cr per km of widening
SCENARIOS = {
    "Scenario 1 (links 7,21,29)": {
        "cost_file": "3_scenario_A_car.omx",
        "links": [(7, 16.63), (21, 16.30), (29, 10.27)]
    },
    "Scenario 2 (links 7,21,40)": {
        "cost_file": "3_scenario_B_car.omx",
```

```

"links": [(7, 16.63), (21, 16.30), (40, 9.66)]},
"Scenario 3 (links 29,40,25)": {
"cost_file": "3_scenario_C_car.omx",
"links": [(29, 10.27), (40, 9.66), (25, 9.78)]
},
"Scenario 4 (links 33,35,40)": {
"cost_file": "2_scenario_D_car.omx",
"links": [(33, 3.14), (35, 9.18), (40, 9.66)]
},
}
def calculate_vht(trips_path: str, cost_path: str) -> float | None:
"""Compute VHT =  $\sum(T_{ij} \times C_{ij}) / 60$ , returns None on error."""
try:
with omx.open_file(trips_path, 'r') as t, omx.open_file(cost_path, 'r') as c:
trips = np.array(t[t.list_matrices()[0]])
costs = np.array(c[c.list_matrices()[0]])
return float(np.sum(trips * costs) / 60.0)
except Exception as e: print(f" [ERROR reading {os.path.basename(cost_path)}]: {e}")
return None
def widening_cost_cr(links: list[tuple]) -> float:
"""Total widening cost in ₹ Crore."""
return sum(length_km * WIDEN_COST_CR for _, length_km in links)
def annual_saving_cr(daily_vht_saving: float) -> float:
"""Annual travel cost saving in ₹ Crore."""
return daily_vht_saving * VHT_COST_RS * 365 / 1e7
def roi(annual_cr: float, widen_cr: float) -> float:
"""ROI ratio = annual saving / widening cost."""
return annual_cr / widen_cr if widen_cr > 0 else float('nan')
def payback_years(annual_cr: float, widen_cr: float) -> str:
"""Payback period in years (or 'N/A' if saving ≤ 0)."""
if annual_cr <= 0:

```

```

return "N/A"
return f"{widen_cr / annual_cr:,.0f}" * 100
print(f"TASK 4 — MULTI-SCENARIO VHT & ROI REPORT: ^100}")
print("=" * 100)
print(f"Base VHT : {BASE_VHT:,.2f} veh·hr/day")
print(f"Base cost/day : ₹{BASE_VHT * VHT_COST_RS:,.2f}")
print(f"VHT rate : ₹{VHT_COST_RS}/veh·hr | Widening rate: ₹{WIDEN_COST_CR} Cr/km")
print("=" * 100)
results = []
for name, cfg in SCENARIOS.items():
    full_path = os.path.join(FOLDER, cfg["cost_file"])
    if not os.path.exists(full_path):
        print(f"\n [MISSING] {cfg['cost_file']} — skipping {name}")
        continue
    vht = calculate_vht(TRIPS_FILE, full_path)
    if vht is None: continue
    saving = BASE_VHT - vht
    daily_rs = saving * VHT_COST_RS
    ann_cr = annual_saving_cr(saving)
    wc = widening_cost_cr(cfg["links"])
    r = roi(ann_cr, wc)
    pay = payback_years(ann_cr, wc)
    link_str = ", ".join(f"{lid}({km}km)" for lid, km in cfg["links"])
    results.append({
        "Scenario" : name,
        "Links (length)" : link_str,
        "New VHT" : vht,
        "VHT Saving" : saving,
        "% Reduction" : saving / BASE_VHT * 100,
        "Daily Saving (₹)" : daily_rs,
        "Annual Saving (Cr)": ann_cr,
    })

```

```

"Widen Cost (Cr)" : wc,
"ROI ratio" : r,"Payback (yrs)" : pay,
})
if results:
col_w = [32, 30, 12, 14, 10, 18, 20, 18, 12, 15]
headers = ["Scenario","Links (length)","New VHT","VHT Saving",
"% Red.","Daily Saving (₹)","Annual Saving (₹Cr)","Widen Cost (₹Cr)","ROI","Payback (yrs)"]
sep = "—" * sum(col_w)
print()
print(sep)
header_line = "".join(f"{h:<{w}} " for h, w in zip(headers, col_w))
print(header_line)
print(sep)
for r in results:
saving = r["VHT Saving"]
flag = " ◀ BEST" if r == max(results, key=lambda x: x["ROI ratio"]) else ""
braess = " ← BRAESS PARADOX" if saving < 0 else ""line = (
f"{r['Scenario']:<{col_w[0]}} "
f"{r['Links (length)']:<{col_w[1]}} "
f"{r['New VHT']:>{col_w[2]-1}.2f} "
f"{saving:>{col_w[3]-1}.2f} "
f"{r['% Reduction']:>{col_w[4]-1}.2f} %"
f"{r['Daily Saving (₹)']:>{col_w[5]-1}.2f} "
f"{r['Annual Saving (Cr)']:>{col_w[6]-1}.5f} "
f"{r['Widen Cost (Cr)']:>{col_w[7]-1}.1f} "
f"{r['ROI ratio']:>{col_w[8]-1}.6f} "
f"{r['Payback (yrs)']:>{col_w[9]-1}} "
f"{flag} {braess}"
)
print(line)
print(sep)

```

```
best = max(results, key=lambda x: x["ROI ratio"])
print(f"\n ★ Best ROI : {best['Scenario']}")print(f" VHT saved : {best['VHT Saving']:.2f} veh·hr/day ({best['% Reduction']:.2f}% reduction)")
print(f" Daily saving : ₹{best['Daily Saving (₹)']:.2f} ")
print(f" Annual saving : ₹{best['Annual Saving (Cr)']:.5f} Cr")
print(f" Widening cost : ₹{best['Widen Cost (Cr)']:.1f} Cr")
print(f" ROI ratio : {best['ROI ratio']:.6f} ")
print(f" Payback period: {best['Payback (yrs)']} years")
df = pd.DataFrame(results)
out_csv = os.path.join(FOLDER, "task4_scenario_results.csv")
df.to_csv(out_csv, index=False)
print(f"\n Results saved to: {out_csv}")
print("\n" + "=" * 100 + "\n")
```